

Henry Lee

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EDUCATION

Purdue University | West Lafayette, IN

GPA: 3.6/4.0

Bachelor of Science, Major in Computer Science, Minor in Mathematics

August 2023 – Present

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, C Programming, Object-Oriented Programming, Systems Programming, Algorithms Analysis, Introduction to Artificial Intelligence, Multivariate Calculus, Linear Algebra, Statistics for Data Science

EXPERIENCE

Machine Learning Co-Op

August 2025 – December 2025

Purdue University: The Data Mine (Empire Cooler)

West Lafayette, IN

- Built a Python tool to help technicians quickly find accurate repair information
- Automated web-scraping and data cleaning to create a structured technical dataset
- Worked in an Agile team to design and refine an AI-powered prototype

Software Engineering Intern

October 2024 – August 2025

Echopath LLC

Indianapolis, IN

- Automated financial and ticket workflows with Python to improve work speed
- Optimized SQL databases with indexing and normalization, improving query speeds for 700K+ records
- Delivered internal engineering tools that streamlined data access and increased operational efficiency
- Utilized Python to build internal engineering tools with Tkinter, Rust, and SQL pipelines

Software Engineering Co-Op

August 2024 – December 2024

Purdue University: The Data Mine (Concrete Engine)

West Lafayette, IN

- Collaborated in a startup to assist in developing a high-performance console program for AI/HPC workflows
- Worked in an Agile structure to develop a customer console program and build MongoDB models
- Researched various possible feature implementations, such as Google Cloud Storage and authentication

Data Science Co-Op

January 2024 – May 2024

Purdue University: The Data Mine (Wikimedia Deutschland)

West Lafayette, IN

- Detected 900+ data mismatches by comparing mismatches to improve data quality
- Improved dataset reliability and security by integrating REST APIs for automated attribute validation
- Constructed a Python and SPARQL pipeline to retrieve data from Wikidata and external data sources

PROJECTS

Multilingual Healthcare Communication App | React, Python, APIs

December 2025 - Present

- Building a real-time multilingual medical translation app for patient-provider communication
- Integrating speech-to-text and neural translation for accurate medical terminology

Ticket Search Utility | Python, Rust, Flask, SQL, APIs

April 2025 – August 2025

- Developed a ticket search application to search for specific ticket notes at Echopath LLC
- Enhanced search accuracy through advanced filters and SQL query optimization to reduce search time
- Engineered a full-stack Flask and SQL platform of 700K+ tickets collected from ConnectWise API

Tickets@Purdue | Java, Swing, JUnit, Git, GitHub, IntelliJ

October 2023 – December 2023

- Built a secure ticket marketplace for Purdue students with real-time Java transactions
- Designed a Swing GUI and JUnit tests for usability and reliability

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Rust, JavaScript, TypeScript, SQL, MATLAB, C#

Libraries/Frameworks: Flask, React, Jupyter, JUnit, Node.js, Next.js, scikit-learn, numpy, pandas, matplotlib

Developer Tools: Git, VS Code, PyCharm, IntelliJ, Google Colab

ENGAGEMENT

Poster Presentations: The Data Mine 2024, Purdue Spring Symposium 2025, Purdue Summer Symposium 2025

Hobbies: Digital Art, Writing, Volunteering